



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electronics and Telecommunication Engineering
Summer Term Examination
Academic Year: 2024-25

Class: EY B.Tech	Subject: Numerical Methods	Subject Code: ESC283
Date: 25.06.2025	Time: 14:30 to 17:30	Max. Marks: 100

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iii. All questions are compulsory.

iii. All questions are compulsory.

Q.N.	Questions	Marks	CO	BT																				
Q.1	1. Explain accuracy and precision with proper examples. 2. Define an error. Also explain various types of errors.	10	CO1	BT1																				
Q.2	Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by using i) Trapezoidal rule ii) Simpson's (1/3) rd rule iii) Simpson's (3/8) th rule	15	CO3	BT5																				
Q.3	Find the root of the equation $\cos x = xe^x$ using the bisection method correct to three decimal places.	15	CO2	BT3																				
Q.4	Apply Gauss-Seidal method iteration method to solve the equations $20x + y - 2z = 17$; $3x + 20y - z = -18$; $2x - 3y + 20z = 25$	10	CO5	BT4																				
Q.5	Using Euler's method, find an approximate value of y corresponding to x=1 Given that $\frac{dy}{dx} = x + y$ and $y = 1$ when $x = 0$.	15	CO1	BT1																				
Q.6	Fit a straight line to the following data: <table><tr><td>x</td><td>6</td><td>7</td><td>7</td><td>8</td><td>8</td><td>8</td><td>9</td><td>9</td><td>10</td></tr><tr><td>y</td><td>5</td><td>5</td><td>4</td><td>5</td><td>4</td><td>3</td><td>4</td><td>3</td><td>3</td></tr></table>	x	6	7	7	8	8	8	9	9	10	y	5	5	4	5	4	3	4	3	3	10	CO3	BT2
x	6	7	7	8	8	8	9	9	10															
y	5	5	4	5	4	3	4	3	3															
Q.7	Use Romberg's method to compute $\int_0^1 \frac{dx}{1+x^2}$ correct to 4 decimal places.	10	CO3	BT5																				

Q.8	Using the regula falsi method, find the real root of the equation $x \log_{10} x = 1.2$ correct to four decimal places	15	CO4	BT
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=== ALL THE BEST ===

Bloom's Taxonomy (BT):

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes: On successful completion of this course, students should be able to:

CO 1	To Understand Accuracy and precision with examples.
CO 2	To locating roots of the equations using Graphical method, Bisection method and false Position method.
CO 3	Understand one point iterative method to find True roots.
CO 4	To Know the Open method like Newton Rapson's Method, Multiple Roots and Secant method.
CO 5	To find unknowns using Gauss Elimination method, Gauss Jordan method and Gauss Seidel method.



**Shri Guru Gobind Singhji Institute of Engineering and Technology,
Nanded**

Department of BSH&M (Mathematics)

Summer term Examination Academic Year: 2024-25

Class: SY B.Tech.	Subject: M-IV (Statistical methods & Complex analysis)	Subject Code: BSC275
Date: 16/06/2025	Time: 14.30-17.30	Max. Marks: 100

Note: i) Nonprogrammable Calculator is allowed, ii) This paper contains total 3 page.

Q.1	Attempt any five	Marks	CO	BT
a	9 dissimilar balls are distributed at random into 3 boxes marked A, B, C. Find the probability that these boxes contain respectively 3, 4, 2 balls.	06	CO5	BT2
b	Five men in a company of 20 are graduates. If three men are picked out of the 20 at random then i) what is the probability that they are all graduates? ii) what is the probability of exactly one is graduate?	06	CO5	BT2
c	A coin is tossed 10 times. Find the probability of getting (i) Exactly 8 heads (ii) 7 heads and 3 tail (iii) At most 2 heads.	06	CO5	BT2
d	Find the principal argument & polar form of complex number $z = \frac{-\sqrt{3}}{2} + \frac{1}{2}i$	06	CO1	BT1
e	Find all the 4 th roots of $z = 1 - \sqrt{3}i$.	06	CO1	BT1
f	i) Find the points at which the function $f(z) = e^{-y}(-\sin x + i \cos x)$ is analytic.	06	CO1	BT2
Q.2	Attempt any six			
a	Three boxes I, II and III, contain balls as follows: • Box I contains 4 white & 2 black balls. • Box II contains 1 white & 3 black balls. • Box III contains 4 white & 3 black balls. A box is chosen at random, and a ball is drawn out of it. If the ball is found to be black, what is the probability that Box II was selected.	08	CO5	BT2
b	A box contains 4 white and 5 black balls. If 3 balls are drawn at random, Find the mathematical expectation of the number of white balls.	08	CO5	BT3
c	Given that $P(A) = \frac{3}{8}$, $P(B) = \frac{5}{8}$ & $P(A \cup B) = \frac{3}{4}$, Find the probabilities: $P(\bar{A})$, $P(A \cap B)$, $P(A B)$, $P(\bar{A} \cap B)$.	08	CO5	BT1
d	The p.d.f. of a continuous random variable X is $f(x) = k(-x^2 + 3x - 2)$; $(1 \leq x \leq 2)$. Determine, i) The value of the constant k. ii) The cumulative distribution function. iii) The probability that X is less than $\frac{1}{2}$.	08	CO5	BT2

e	The following is the table which gives you the distribution of marks secured by some students in an examination. Find the median for the following data.						08	CO5	BT1		
	Marks	10-20	20-30	30-40	40-50	50-60					
	No. of students	42	85	120	84	49					
f	Find the Mean and mode of the following distribution						08	CO5	BT1		
	Class	20-30	30-40	40-50	50-60	60-70					
	Frequency	8	7	20	5	10					
g	i) Evaluate $\int_C \frac{z^2 e^z}{(z-2)^2} dz$ where C is the circle of radius 4 with centre at 0 oriented anticlockwise.						06	CO3	BT3		
	ii) Find $\int_C \frac{\cos(z^2+3i)}{e^z} dz$, where C is the rectangle with vertices $1+i$, $-1+i$, $-1-i$, $1-i$ with anticlockwise orientation.										
Q.3 Attempt any three											
a	For two events A and B, Let $P(A) = 0.2$, $P(A \cup B) = 0.5$ and $P(B) = p$.						06	CO5	BT2		
	(i) For what value of p, A and B are independent?										
	(ii) For what value of p, events A and B are mutually exclusive?										
b	i) An unbiased die is thrown. What is the mathematical expectation of the number of points?						06	CO5	BT2		
	ii) If 4 dice are thrown simultaneously, find the expected sum of the points on the upper surfaces of the dice.										
c	A discrete random variable X has the following probability mass function:						06	CO5	BT2		
	x	0	1	2	3	4				5	6
	f(x)	k	2k	3k	4k	5k				6k	7k
d	i) Find value of k. ii) Find $P(X < 4)$, $P(X \geq 5)$ & $P(3 < X \leq 6)$.						06	CO1	BT3		
	i) Find the principal logarithm ($\text{Ln}Z$) for the complex number $Z = \sqrt{3} - i$.										
	ii) Find the principal value of $(1+i)^{1-i}$ (Write solution in detail).										
Q.4 Attempt any one											
a	The marks obtained by 10 students are:						04	CO5	BT1		
	Marks (x)	2	4	6	8	10					
	Frequency (f)	1	2	3	2	2					
b	Find the Mean ($\bar{x}$) & standard deviation(σ) for the above data.						04	CO1	BT2		
	Determine the value of "a" for which the function										
	$u(x, y) = e^{3x} \cos(ay)$ is harmonic.										

-----All the best-----

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create.

Course Outcomes:

On successful completion of this course, students should be able to:

	SGGS Institute of Engineering and Technology, Vishnupuri, Nanded.	
	Summer -Term Examination 2024-25	
	S.Y. B. Tech.	Department: Mathematics
	Mathematics: IV (Statistical and Numerical methods)	BSC - 274
Date: 16/06/2025	Time: 14:30-17:30	Max. Marks: 100

Note:

- (i) Attempt all questions
- (ii). Assume suitable data if necessary
- (iii). Use of a non-programmable calculator is permitted.
- (iv). This paper contains 02 pages.

Q1. Attempt any five.		Marks	CO& BT																												
(a)	(i) Find out the missing frequencies of the following data, given that Arithmetic Mean is 67.45 inches. <table><tr><td>Height(inches)</td><td>60-62</td><td>63-65</td><td>66-68</td><td>69-71</td><td>72-74</td><td>Total</td></tr><tr><td>No.of students</td><td>6</td><td>17</td><td>f_3</td><td>f_4</td><td>8</td><td>100</td></tr></table> (ii)Find the modal values for the monthly profits in rupees of 100 shops are distributed as follows <table><tr><td>Profits per shops</td><td>0-100</td><td>100-200</td><td>200-300</td><td>300-400</td><td>400-500</td><td>500-600</td></tr><tr><td>No. of Shops</td><td>12</td><td>18</td><td>27</td><td>20</td><td>17</td><td>6</td></tr></table>	Height(inches)	60-62	63-65	66-68	69-71	72-74	Total	No.of students	6	17	f_3	f_4	8	100	Profits per shops	0-100	100-200	200-300	300-400	400-500	500-600	No. of Shops	12	18	27	20	17	6	08	CO2 BT5
Height(inches)	60-62	63-65	66-68	69-71	72-74	Total																									
No.of students	6	17	f_3	f_4	8	100																									
Profits per shops	0-100	100-200	200-300	300-400	400-500	500-600																									
No. of Shops	12	18	27	20	17	6																									
(b)	Three identical boxes I, II, III contain respectively 3 white & 2 red balls, 2 white & 6 red balls, and 1 white & 2 red balls. A box is chosen at random, and a ball is drawn out of it. If the ball is found to be white, what is the probability that Box II was selected.	08	CO2 BT2 BT3																												
(c)	If A and B are independent events, and $P(A) = \frac{3}{4}$, $P(B) = \frac{4}{5}$, then i) Find values of the following probability $P(\bar{A})$, $P(\bar{B})$, $P(A + B)$, $P\left(\frac{A}{B}\right)$, and $P(\bar{A}B)$.	08	CO2 BT3																												
(d)	The overall percentage of failures in a certain examination is 50. What is the probability that out of a group of 8 candidates at least 4 passed the examination. Also find mean and variance.	08	CO2 BT2 BT3																												
(e)	A random variable x follows Poisson distribution with parameter 4, Find the probabilities that the variable assumes the values (i) 0, 1, 2, 3, 4; (ii) Less than 4 (iii) At least 2.	08	CO1 BT3																												
(f)	A subcommittee of 8 members is to be formed out of a group consisting of 9 men and 4 ladies. Calculate the probability that the subcommittee will consist of (i) Exactly 2 ladies, (ii) At least 2 ladies, (iii) Exactly 3 ladies, (iv) At least 3 ladies.	08	CO2 BT2																												

Q2. Attempt any five																					
(a)	Using Newton - Raphson method find the root between 4 and 5 of $x^2 - 5x + 2 = 0$ correct to 4 decimal places.	08	CO1 BT1																		
(b)	Find a real root between 1 and 1.5 of the equation $xe^x - 3 = 0$, using Regula-Falsi method correct to three decimal places.	08	CO1 BT5																		
(c)	Find the value of $\sin 52^\circ$ from the given table. <table><tr><td>θ</td><td>45°</td><td>50°</td><td>55°</td><td>60°</td></tr><tr><td>$\sin \theta$</td><td>0.7071</td><td>0.7660</td><td>0.8192</td><td>0.8660</td></tr></table>	θ	45°	50°	55°	60°	$\sin \theta$	0.7071	0.7660	0.8192	0.8660	08	CO1 BT2								
θ	45°	50°	55°	60°																	
$\sin \theta$	0.7071	0.7660	0.8192	0.8660																	
(d)	Using the following data, find $f(7.5)$. <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>$f(x)$</td><td>1</td><td>8</td><td>27</td><td>64</td><td>125</td><td>216</td><td>343</td><td>512</td></tr></table>	x	1	2	3	4	5	6	7	8	$f(x)$	1	8	27	64	125	216	343	512	08	CO2 BT3
x	1	2	3	4	5	6	7	8													
$f(x)$	1	8	27	64	125	216	343	512													
(e)	A random variable X having the following probability mass function <table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>$P(X = x)$</td><td>0</td><td>k</td><td>$2k$</td><td>$2k$</td><td>$3k$</td><td>k^2</td><td>$2k^2$</td><td>$7k^2 + k$</td></tr></table> Find (i) Determine constant k, (ii) Find $P(X < 6)$, (iii) Find $P(X \geq 6)$, (iv) Find $P\left(\frac{X < 6}{3 < X \leq 7}\right)$, (v) Find $P\left(\frac{X=4}{4 \leq X \leq 7}\right)$.	X	0	1	2	3	4	5	6	7	$P(X = x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$	08	CO2 BT2
X	0	1	2	3	4	5	6	7													
$P(X = x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$													
(f)	The p.d.f. of continuous random variable is $f(x) = k(x - 1)(2 - x)$, $1 \leq x \leq 2$, determine (i) The value of the constant k; (ii) The cumulative distribution function. (iii) The probability that X is less than $\frac{5}{4}$; (iv) the probability that X is greater than $\frac{3}{2}$; (v) Probability that X lies between $\frac{5}{4}$ and $\frac{3}{2}$.	08	CO2 BT2																		
Q3. Attempt any four																					
(a)	A bag contains 7 red and 6 white balls. Two successive draws of 3 balls are made without replacement. Find the probability that the first drawing will give 3 white balls and the second 3 red balls.	05	CO2 BT2																		
(b)	An urn contains 8 white and 3 red balls. If 2 balls are drawn at random, find the probability that (i) Both are white, (ii) both are red, (iii) one each of color.	05	CO2 BT2																		
(c)	A box contains 4 white and 6 black balls. If 3 balls are drawn at random, find the mathematical expectation of the number of white balls.	05	CO2 BT2																		
(d)	Find the Median and Median class of the data given below. <table><tr><td>Class</td><td>15-25</td><td>25-35</td><td>35-45</td><td>45-55</td><td>55-65</td><td>65-75</td></tr><tr><td>Frequency</td><td>4</td><td>11</td><td>19</td><td>14</td><td>0</td><td>2</td></tr></table>	Class	15-25	25-35	35-45	45-55	55-65	65-75	Frequency	4	11	19	14	0	2	05	CO2 BT2				
Class	15-25	25-35	35-45	45-55	55-65	65-75															
Frequency	4	11	19	14	0	2															
(e)	A coin is tossed 10 times. Find the probability of getting (i) Exactly 6 heads, and (ii) 9 heads and 1 tail.	05	CO2 BT2																		

[BT1: Remembering, BT2: Understanding, BT3: Applying, BT4: Analysing, BT5: Evaluating, BT6: Creating]
CO1. To develop techniques of data interpretation.

CO2. Develop problem solving techniques needed to accurately calculate probabilities and describe the properties of discrete and continuous distribution functions.

CO3. Use statistical tests in testing hypotheses on data, compute covariances, and correlations, Apply the tests of goodness of fit.

CO4. Develop the numerical skills for finding roots of polynomial and transcendental equations.

CO5. Conduct numerical integration and differentiation and solve ODE's and PDE's and engineering problems.



Shri Guru Gobind Singhji Institute of Engineering
and Technology, Vishnupuri, Nanded- 431606.

Exam: Summer Term Examination (2024-25)

Class: S. Y. B. Tech (CSE, INFT) Department: Mathematics

Course Code and Name: BSC276, M-IV Vector Calculus and Statistical Methods

Day and Date: Monday, 16/06/2025 Time: 14:30 - 17:30 Maximum Marks: 100

- All questions are compulsory.
- Calculators are allowed.

Q 1) Solve any two of the following. (10 marks) [CO1]

1. Find the work done by the force field $F(x, y, z) = xi + yj + zk$ when a particle is moved along the straight-line segment from $(1, 2, 1)$ to $(4, 5, 3)$. [BT2, BT3, BT5]
2. Evaluate the line integral $\int_C x^2 dx + xy dy + dz$, where C is parameterized by $c(t) = (t, t^2, 1)$, $0 \leq t \leq 1$. [BT2, BT3, BT5]
3. Let D be the rectangle in the $\theta\phi$ plane defined by $0 \leq \theta \leq 2\pi$, $0 \leq \phi \leq \pi$, and let the surface S be defined by the parametrization $\Phi: D \rightarrow \mathbb{R}^3$ is given by $x = \cos\theta\sin\phi$, $y = \sin\theta\sin\phi$, $z = \cos\phi$. Here θ and ϕ are the angles of spherical coordinates, and S is the unit sphere parameterized by Φ . Let $\mathbf{r}$ be the position vector $\mathbf{r}(x, y, z) = xi + yj + zk$. Compute $\iint_S \mathbf{r} \cdot d\mathbf{S}$. [BT2, BT3, BT5]

Q 2) Solve the following. (30 marks) [CO2]

1. a) How many different letter arrangements can be formed from the letters of word "arrangement"?
b) From a group of 6 women and 8 men, how many different committees consisting of 3 women and 2 men can be formed?
c) How many 4 card hands can be formed from standard 52-card deck such that hand containing 3, but not 4, Aces?
d) Customers are allowed to choose tray of 20 number of fruits from 12 variety of fruits. What are the number of possible combinations to choose 20 fruits? [BT5]
2. a) A box contains 6 red, 4 white and 5 black balls. A person draws 4 balls from the box at random. Find the probability that among the balls drawn there is at least one ball of each color. [BT4, BT5]
b) An urn contains 4 tickets numbered 1, 2, 3, 4 and another contains 6 tickets numbered 2, 4, 6, 7, 8, 9. If one of the two urns is chosen at random and a ticket is drawn from the chosen urn, find the probabilities that the ticket drawn bears the number 2 or 4. [BT4, BT5]
3. Suppose someone rents movies from two different DVD stores A and B. Sometimes it happens that the DVD does not work because of scratches. The probability of the DVD working without any problems is 0.90 if it is rented from store A and 0.85 if it is rented from store B. What is the probability that a DVD is from store A given that it is not working? [BT3, BT5]
4. How many arrangements of 5α , 5β and 5γ are there with at least one β and at least one γ between each pair of α ? [BT2, BT4, BT5]
5. Define the random variable. Define the expectation and variance of a random variable for discrete and continuous variable. Prove that $E(a + bX) = a + bE(X)$ and $\text{Var}(a + bX) = b^2\text{Var}(X)$, where a and b are known constant. [BT1, BT2]

Q 3) Solve any four of the following. (28 marks) [CO3]

1. A random variable X has the following distribution:

x	0	1	2	3	4	5	6	7
$p(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

- (a) Find k ; (b) If $P(X \leq c) > \frac{1}{2}$, find the minimum value of c ; (c) Evaluate $E(X)$. [BT3, BT5]
2. Felix states that he is able to distinguish a freshly ground coffee blend from an ordinary supermarket coffee. One of his friends asks him to taste 10 cups of coffee and tell him which coffee he has tested. Suppose that Felix has actually no clue about coffee and simply guesses the brand. What is the probability of at least 7 correct guesses? [BT4, BT5]
3. A coin is tossed until a head appears. What is the expectation of the number of tosses required [BT3, BT5]
4. A manufacturer of cotter pins knows that 5% of his product is defective. If he sells cotter pins in boxes of 100 and guarantees that not more than 10 pins will be defective, what is the approximate probability that a box will fail to meet the guaranteed quality? [BT3, BT5]
5. An innovative winemaker experiments with new grapes and adds a new wine to his stock. The percentage sold by the end of the season depends on the weather and various other factors. It can be modelled using the random variable X with the CDF as $F(x) = \begin{cases} 0 & \text{if } x < 0; \\ 3x^2 - 2x^3 & \text{if } 0 \leq x \leq 1; \\ 1 & \text{if } 1 > x. \end{cases}$
- (a) Determine probability density function $f(x)$.
 (b) What is the probability of selling at least one-third of his wine, but not more than two thirds?
 (c) What is the variance of X . [BT2, BT3, BT5]

Q 4) Solve the following. (32 marks) [CO3]

1. Define a Poisson distribution. Find the mean and variance of a Poisson distribution. [BT1, BT2]
2. X is a normal variate with mean 30 and standard deviation 5. Find the probabilities that (i) $26 \leq X \leq 40$, (ii) $X \geq 45$, and (iii) $|X - 30| > 5$. Given that the area under the standard normal curve: between $z = 0$ and $z = 0.8$ is 0.2881, between $z = -2$ and $z = 2$ is 0.9544, between $z = 0$ and $z = 3$ is 0.49865, between $z = -1$ and $z = 1$ is 0.6826. [BT3, BT4, BT5]
3. The height distribution of a group of 10,000 men is normal with mean height 64.5" and standard deviation 4.5". Find the number of men whose height is (a) less than 69" but greater than 55.5"; (b) less than 55.5"; (c) more than 73.5". (For standard normal variable Z , the area between $z = -1$ and $z = 1$ is 68.27%, the area between $z = -2$ and $z = 2$ is 95.45%, the area between $z = -3$ and $z = 3$ is 99.73%.) [BT3, BT5]
4. If m things are distributed among a men and b women, show that the probability that the number of things received by men is odd, is $\frac{1}{2} \left[\frac{(b+a)^m - (b-a)^m}{(b+a)^m} \right]$. [BT3, BT5]

Bloom's Taxonomy:

BT1: Remembering BT2: Understanding BT3: Applying
 BT4: Analyzing BT5: Evaluating BT6: Creating

Course Outcomes:

CO1: Evaluate line integrals, surface integrals, and volume integrals and convert line integrals into area integrals and surface integrals into volume integrals using integral theorems.

CO2: To develop techniques of data interpretation.

CO3: Develop problem solving techniques needed to accurately calculate probabilities and describe the properties of discrete and continuous distribution functions.

CO4: Use statistical tests in testing hypotheses on data.

CO5: Compute covariances, and correlations, Apply the tests of goodness of fit.



Shri Gurn Gobind Singhji Institute of Engineering
and Technology, Vishnupuri, Nanded- 431606.

Exam: Summer Term Examination (2024 - 25)

Class: S. Y. B. Tech (CSE, INFT) Department: Mathematics

Course Code and Name: BSC273, M-III : Applied Linear Algebra

Day and Date: Tuesday, 17/06/2025 Time: 14:30 - 17:30 Maximum Marks: 100

1. All questions are compulsory.
2. Calculators are allowed.
3. A vector in $\mathbb{R}^n$ is written as $(x_1, x_2, \dots, x_n)$.

Question 1) Attempt any three of the following. (24 marks)

1. Define a vector space. Let M_{22} be the set of all 2×2 matrices whose entries are real numbers. Prove that M_{22} is a vectors space with usual matrix addition and scalar multiplication, where scalars are real numbers. [CO1, BT1, BT3]
2. Let V and W be two finite-dimensional vector spaces (over same field of scalars). Then V is isomorphic to W if and only of $\dim(V) = \dim(W)$. [CO2, BT1, BT2]
3. Let $\mathbb{R}^n$ be a vector space over field $\mathbb{R}$, where $n \geq 1$. Define $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ as $\langle (x_1, x_2, \dots, x_n), (y_1, y_2, \dots, y_n) \rangle = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$. Prove that $\mathbb{R}^n$ is an inner product space with inner product $\langle \cdot, \cdot \rangle$. Prove that standard basis of $\mathbb{R}^n$ is a orthonormal basis. [CO5, BT2, BT3]
4. Explain in detail a real life application of eigenvalue and eigenvector with an example. [CO4, BT4, BT6]

Question 2) Solve the following. [CO4, BT3, BT5] (36 marks)

1. Find the geometric and algebraic multiplicity of each eigenvalue of matrix $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$, and determine whether A is diagonalizable. If A is diagonalizable, then find a matrix P that diagonalizes A , and find $P^{-1}AP$.
2. Find A^n if n is a positive integer and $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 3 & 0 \\ -1 & 1 & 4 \end{bmatrix}$.
3. Assume that the vector space $\mathbb{R}^4$ has the Euclidean inner product. Apply the Gram-Schmidt process to transform basis vectors $u_1 = (1, 1, 1, 1), u_2 = (0, 1, 1, 1), u_3 = (0, 0, 1, 1), u_4 = (0, 0, 0, 1)$ into an orthogonal basis $\{v_1, v_2, v_3, v_4\}$ and then normalize the orthogonal basis vectors to obtain an orthonormal basis $\{q_1, q_2, q_3, q_4\}$.

Question 3) Solve the following. [CO3, BT5] (15 marks)

1. Let $S = \{(1, 1, 0, 0), (0, 1, 1, 0), (0, 0, 1, 1), (1, 0, 0, 1)\}$ be a subset of $\mathbb{R}^4$. Determine whether the set S is linearly dependent or independent. [BT2, BT3]
2. Determine whether $W = \{A \in M_{nn} \mid AB = BA\}$ is subspace of M_{nn} , where M_{nn} is set of all $n \times n$ real matrices and B is a given fixed matrix. [BT2, BT3]

3. Determine whether $T: M_{nn} \rightarrow \mathbb{R}$ defined by $T(A) = \text{trace}(A)$ is a linear transformation.
4. Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be a linear transformation defined by $T((a, b)) = (a + 2b, -a, b)$ and $B = \{(1, 2), (3, -1)\}$ a basis of $\mathbb{R}^2$, $C = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$ a basis of $\mathbb{R}^3$. Find matrix of T . [CO2, BT5]
5. Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation which rotates a vector v by an angle θ in anticlockwise direction, where $0 \leq \theta \leq 90$. Find the matrix A of T with respect to a standard basis.

Question 4) Solve the following. [CO3] (25 marks)

1. Let $T: \mathbb{P}_2 \rightarrow M_{22}$ be a linear transformation defined by $T(a+bx+cx^2) = \begin{bmatrix} a+b & a+2c \\ 2a+c & b-c \end{bmatrix}$. Determine whether the linear transformation T is (a) one-to-one (b) onto. [BT2, BT5]
2. Let $S = \{(1, 1, 0, \dots, 0), (0, 1, 1, 0, \dots, 0), (0, 0, 1, 1, 0, \dots, 0), \dots, (1, 0, \dots, 0, 1)\}$ be a subset of $\mathbb{R}^n$, where $n \geq 3$. Determine whether the set S is linearly dependent or independent. [BT2, BT3]
3. Define matrix norm and explain the one-norm with an example. [BT1, BT2]
4. Compute $\begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}^{10} \begin{bmatrix} 5 \\ 1 \end{bmatrix}$. [BT3, BT5]
5. Let $\{v_1, v_2, \dots, v_n\}$ be a basis for a vector space V . Prove that $\{v_1, v_1 + v_2, v_1 + v_2 + v_3, \dots, v_1 + v_2 + \dots + v_n\}$ is also basis for V . [BT2, BT3]

Bloom's Taxonomy:

BT1: Remembering	BT2: Understanding	BT3: Applying
BT4: Analyzing	BT5: Evaluating	BT6: Creating

Course Outcomes:

- CO1: To understand fields and vector spaces, subspaces, linear independence and dependence.
- CO2: To find basis and dimension of a vector space and understand change of basis. Find a basis for the row space, column space and null space of a matrix and find the rank and nullity of a matrix.
- CO3: To compute linear transformations, kernel and range, and inverse linear transformations, and find matrices of general linear transformations.
- CO4: To understand eigenvalues and eigenvectors and diagonalization process.
- CO5: To learn inner products on a real vector space and orthogonality in inner product spaces and obtain orthonormal bases using Gram-Schmidt process.
- CO6: To learn the different matrix norms, convergence of matrices and matrix decompositions such as QR, SVD, LU, Cholesky.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of BSH&M (Mathematics)
Summer term Examination Academic Year: 2024-25 (Semester – I)

Class: SY B.Tech.	Subject: M-III (Complex analysis, Probability and Statistics).	Subject Code: BSC203
Date: 17/06/2025	Time: 14.30-17.30	Max. Marks: 100

Note: i) Nonprogrammable Calculator is allowed, ii) This paper contains total 2 page.

Q.1	Attempt any five	Marks	CO	BT
a	i) Find all the 4 th roots of $z = -1 + \sqrt{3}i$. ii) Sketch the set of solutions of following inequality $ Arg(z) \leq \frac{\pi}{4}$.	06	CO1	BT1
b	i) Find modulus and argument of $z = (1 + \sqrt{3}i)^{30}$. ii) Find $Re(f)$ & $Im(f)$ for $f(z) = \frac{z}{z+1}$.	06	CO1	BT3
c	Find the points at which the function $f(z) = e^{-y}(\cos x + i \sin x)$ is analytic.	06	CO1	BT2
d	Find harmonic conjugate of the function $u(x, y) = x^2 - y^2 - x$.	06	CO1	BT2
e	i) Find the principal logarithm (Lnz) for the complex number $z = \sqrt{3} - i$. ii) Find the principal value of $(1 - i)^{1+i}$ (Write solution in detail).	06	CO1	BT1
f	i) Find all the points at which the function $f(z) = z(z^4 - 5)$ is not conformal. ii) Find the fixed points of following function $f(z) = \frac{2z-5}{z+4}$.	06	CO2	BT2
Q.2	Attempt any five			
a	i) Evaluate $\int_C \frac{z^3 e^z}{(z-2)^2} dz$ where C is the circle of radius 5 with centre at 0 oriented anticlockwise. ii) Find $\int_C \frac{\cos(z^2+3)}{e^z} dz$, where C is the triangle with vertices i, -1-i & 1-i with anticlockwise orientation.	06	CO3	BT3
b	Evaluate $\int_C (2x + 1 + iy) dz$, where C is the path given by $c(t) = (t + 1) + i(2t^2 - 1), 0 \leq t \leq 1$.	06	CO3	BT2
c	10 dissimilar balls are distributed at random into 4 boxes marked A, B, C, D. Find the probability that these boxes contain respectively 4,3,2,1 balls.	06	CO5	BT2
d	Five men in a company of 15 are graduates. If three men are picked out of the 15 at random then i) what is the probability that they are all graduates? ii) what is the probability of exactly one is graduate?	06	CO5	BT2
e	A coin is tossed 10 times. Find the probability of getting (i) Exactly 7 heads (ii) 3 heads and 7 tail (iii) At most 2 heads.	06	CO5	BT2

f	Three boxes I, II and III, contain balls as follows: - Box I contains 2 white & 4 black balls. - Box II contains 3 white & 1 black balls. - Box III contains 2 white & 3 black balls. A box is chosen at random, and a ball is drawn out of it. If the ball is found to be black, what is the probability that Box III was selected.	06	CO5	BT1																
Q.3 Attempt any five																				
a	A box contains 6 white and 4 black balls. If 3 balls are drawn at random, Find the mathematical expectation of the number of white balls.	06	CO5	BT2																
b	Given that $P(A) = \frac{5}{8}$, $P(B) = \frac{3}{8}$ & $P(A \cup B) = \frac{3}{4}$, Find the probabilities: $P(\bar{A})$, $P(A \cap B)$, $P(A B)$, $P(\bar{A} \cap B)$.	06	CO5	BT1																
c	The p.d.f. of a continuous random variable X is $y = k(-x^2 + 3x - 2); (1 \leq x \leq 2).$ Determine, i) The value of the constant k. ii) The cumulative distribution function. iii) The probability that X is less than $\frac{3}{2}$.	06	CO5	BT2																
d	The following is the table which gives you the distribution of marks secured by some students in an examination. Find the median for the following data. <table border="1"> <tr> <td>Marks</td> <td>10-20</td> <td>20-30</td> <td>30-40</td> <td>40-50</td> <td>50-60</td> </tr> <tr> <td>No. of students</td> <td>42</td> <td>85</td> <td>120</td> <td>84</td> <td>49</td> </tr> </table>	Marks	10-20	20-30	30-40	40-50	50-60	No. of students	42	85	120	84	49	06	CO5	BT2				
Marks	10-20	20-30	30-40	40-50	50-60															
No. of students	42	85	120	84	49															
e	Find the Mean and mode of the following distribution <table border="1"> <tr> <td>Class</td> <td>20-30</td> <td>30-40</td> <td>40-50</td> <td>50-60</td> <td>60-70</td> </tr> <tr> <td>Frequency</td> <td>7</td> <td>20</td> <td>8</td> <td>10</td> <td>5</td> </tr> </table>	Class	20-30	30-40	40-50	50-60	60-70	Frequency	7	20	8	10	5	06	CO5	BT1				
Class	20-30	30-40	40-50	50-60	60-70															
Frequency	7	20	8	10	5															
f	i) Determine the value of "a" for which the function $u(x, y) = e^{3x} \cos(ay)$ is harmonic. ii) If $z = \cos(3\pi i)$ then find $Re(z)$.	06	CO1	BT2																
Q.4 Attempt all																				
a	The marks obtained by 10 students are: <table border="1"> <tr> <td>Marks (x)</td> <td>2</td> <td>4</td> <td>6</td> <td>8</td> <td>10</td> </tr> <tr> <td>Frequency (f)</td> <td>1</td> <td>2</td> <td>3</td> <td>2</td> <td>2</td> </tr> </table> Find the Mean ($\bar{x}$) & standard deviation(σ) for the above data.	Marks (x)	2	4	6	8	10	Frequency (f)	1	2	3	2	2	06	CO1	BT1				
Marks (x)	2	4	6	8	10															
Frequency (f)	1	2	3	2	2															
b	A discrete random variable X has the following probability mass function: <table border="1"> <tr> <td>x</td> <td>0</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> <td>6</td> </tr> <tr> <td>f(x)</td> <td>k</td> <td>3k</td> <td>5k</td> <td>7k</td> <td>9k</td> <td>11k</td> <td>13k</td> </tr> </table> i) Find value of k ii) Find $P(X < 4)$, $P(X \geq 5)$ & $P(3 < X \leq 6)$.	x	0	1	2	3	4	5	6	f(x)	k	3k	5k	7k	9k	11k	13k	04	CO1	BT2
x	0	1	2	3	4	5	6													
f(x)	k	3k	5k	7k	9k	11k	13k													

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

	On successful completion of this course, students should be able to:
CO 1	Determine whether a given function is analytic and apply analyticity on harmonic functions and conjugates of harmonic functions.
CO 2	Transform a region to another region using conformal mapping.
CO 3	To evaluate contour integrals using Cauchy's integral theorem and formula.
CO 4	Represent functions as Taylor, power, and Laurent series, classify singularities and poles, find residues and evaluate complex integrals using the residue theorem.
CO 5	Develop problem solving techniques needed to accurately calculate probabilities and describe the properties of discrete and continuous distribution functions.



**Shri Guru Gobind Singhji Institute of Engineering and Technology,
Nanded**

Department of BSH&M (Mathematics)

Summer term Examination Academic Year: 2024-25

Class: SY B.Tech.	Subject: M-III (Numerical methods and Differential equation).	Subject Code: BSC272
Date: 17.06.2025	Time: 14.30-17.30	Max. Marks: 100

Note: i) Nonprogrammable Calculator is allowed, ii) This paper contains total two pages.

Q.1	Attempt any six		Marks	CO	BT												
	a	Using Bisection method determine a real root of the equation $f(x) = x^3 - x - 1 = 0$ lying between 1 and 1.5 (correct to two places of decimal).	07	CO2	BT3												
	b	Find the real root of the equation $xe^x - 3 = 0$ lying between 1 & 2, using regula falsi method correct to three decimal places.	07	CO2	BT3												
	c	By using Newton-Raphson's Method, find the real root of the equation $x^4 - x - 10 = 0$, correct to three places of decimals (you can take initial approximation as $x_0 = 2$).	07	CO2	BT3												
	d	Prove that, i) $\nabla = 1 - E^{-1}$ ii) $\delta = E^{\frac{1}{2}} - E^{-\frac{1}{2}}$ iii) $E\nabla = \Delta$ iv) $\Delta = \delta E^{\frac{1}{2}}$	07	CO2	BT2												
	e	Find $f(0.3)$ from the following given data: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>$f(x)$</td><td>3</td><td>6</td><td>11</td><td>18</td><td>27</td></tr></table>	x	0	1	2	3	4	$f(x)$	3	6	11	18	27	07	CO2	BT3
	x	0	1	2	3	4											
	$f(x)$	3	6	11	18	27											
f	For a certain town the population is given below. Find the population of 1953 by using Newton backward difference formula. <table><tr><td>Year</td><td>1931</td><td>1941</td><td>1951</td><td>1961</td></tr><tr><td>Populations in thousands</td><td>172</td><td>171.7</td><td>157.2</td><td>183.9</td></tr></table>	Year	1931	1941	1951	1961	Populations in thousands	172	171.7	157.2	183.9	07	CO2	BT3			
Year	1931	1941	1951	1961													
Populations in thousands	172	171.7	157.2	183.9													
g	Find the polynomial $f(x)$ of degree 2 such that $f(0) = 12$, $f(3) = 6$, $f(4) = 8$ by using Lagrange's interpolation formula.	07	CO2	BT3													
Q.2	Attempt any five																
	a	Given $y' = \frac{y-x}{y+x}$ with $y(0) = 1$, find $y(0.4)$ in 4 steps by Euler's method.	06	CO4	BT3												
	b	Evaluate $\int_0^6 \frac{1}{x^2+1} dx$ by using Simpson's $\frac{1}{3}$ rule with $h = 1$.	06	CO3	BT3												
	c	Find $\int_0^2 e^{x^2} dx$ approximately using trapezoidal rule with $h = \frac{1}{3}$.	06	CO3	BT3												

	d	Using Taylor's series, find $y(0.1)$ correct to four decimal places if $y(x)$ satisfies $\frac{dy}{dx} = x - y^2$ & $y(0) = 1$.	06	CO4	BT3										
	e	Given $\frac{dy}{dx} = 1 + y^2$ with $y(0) = 0$, find $y(0.2)$ by using Runge-kutta fourth order formula.	06	CO4	BT3										
	f	Given $\frac{dy}{dx} = x + y$ with initial condition $y(0) = 1$, find $y(0.05)$ by using Euler's Modified method (with $h = 0.05$) correct to two decimal places.	06	CO4	BT3										
Q.3 Attempt all															
	a	Find $\int_0^6 \frac{e^x}{1+x} dx$ approximately using Simpson's $\frac{3}{8}$ rule with $h = 1$.	06	CO3	BT3										
	b	Find the cubic polynomial which takes the following values: <table border="1" style="margin-left: 40px;"> <tr> <td>x</td> <td>0</td> <td>1</td> <td>2</td> <td>3</td> </tr> <tr> <td>$f(x)$</td> <td>1</td> <td>2</td> <td>1</td> <td>10</td> </tr> </table>	x	0	1	2	3	$f(x)$	1	2	1	10	06	CO1	BT3
x	0	1	2	3											
$f(x)$	1	2	1	10											
Q.4 Attempt all															
	a	Find the Fourier series of the periodic function $f(x) = \begin{cases} 2 & \text{if } -\pi \leq x < 0 \\ x & \text{if } 0 \leq x < \pi \end{cases}$ with $f(x + 2\pi) = f(x)$.	08	CO5	BT2										
	b	Find the Fourier cosine series of the periodic function $f(x) = x^2, -\pi < x < \pi$ with $f(x + 2\pi) = f(x)$.	06	CO5	BT2										
	c	Make only the difference table for the following data <table border="1" style="margin-left: 40px;"> <tr> <td>x</td> <td>3</td> <td>4</td> <td>5</td> <td>6</td> </tr> <tr> <td>$f(x)$</td> <td>31</td> <td>69</td> <td>131</td> <td>223</td> </tr> </table>	x	3	4	5	6	$f(x)$	31	69	131	223	02	CO2	BT1
x	3	4	5	6											
$f(x)$	31	69	131	223											

Bloom's Taxonomy (BT): BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Develop the numerical skills for error analysis.
CO 2	Find roots of polynomial and transcendental equations using numerical techniques
CO 3	Evaluate numerical integration and differentiation.
CO 4	To use numerical methods to solve ordinary and partial differential equations and other engineering problems.
CO 5	Develop the skills of solving Partial differential equations using separation of variables and Fourier series.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded.

Summer-Term Examination 2024-25

S.Y. B. Tech.

Department: Mathematics

Paper: Mathematics -III
(Transform Calculus and Differential
Equations)

Course Code: BSC271

Date: 17/06/2025

Time: 14:30-17:30

Max. Marks: 100

Note: (i). Attempt all questions

(ii). Assume suitable data if necessary

(iii). Calculator is not allowed.

(iv). This paper contains 02 pages.

Q1	Attempt any five.	Marks	CO& BT
(a)	Find the Laplace transform of the following functions & show the details of your work. (i) $f(t) = e^{at}$, (ii) $f(t) = \sinh at$.	08	CO1 BT2
(b)	Solve the Shifted data initial value problem by the Laplace transform $y'' + y = 2t$, $y\left(\frac{\pi}{4}\right) = \frac{\pi}{2}$, $y'\left(\frac{\pi}{4}\right) = 2 - \sqrt{2}$	08	CO1 BT3
(c)	Find the solution of the initial value problem by the Laplace transform $y'' + 9y = 10e^{-t}$, $y(0) = 0$, $y'(0) = 0$	08	CO3 BT3
(d)	Write the following function using unit step function and find its Laplace transform. $f(t) = \begin{cases} 2 & \text{if } 0 < t < 1 \\ \frac{1}{2}t^2 & \text{if } 1 < t < \frac{\pi}{2} \\ \cos t & \text{if } t > \frac{\pi}{2} \end{cases}$	08	CO3 BT1
(e)	Find the solution of the initial value problem $y'' + 4y = \delta(t - \pi) - \delta(t - 2\pi)$, $y(0) = 0$, $y'(0) = 1$	08	CO1 BT3
(f)	(i) Let $H(s) = \frac{1}{[(s-a)s]}$. Find $h(t)$. (here $h(t) = (f * g)(t)$) (ii) Find $e^{-t} * e^t$.	08	CO1 BT1
Q2	Attempt any four.		
(a)	Find the Fourier series for the periodic function $f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x, & 0 < x < \pi. \end{cases}$	10	CO2 BT3
(b)	Obtain Fourier cosine and sine series for $f(x) = x$ in the interval $0 \leq x \leq \pi$.	10	CO2 BT2
(c)	Derive Fourier cosine and Fourier sine integrals of $f(x) = e^{-35x}$, where $x > 0$.	10	CO2 BT2

(d)	Find the Fourier cosine and Fourier sine transform of the function $f(x) = \begin{cases} k & \text{if } 0 < x < a \\ 0 & \text{if } x > a. \end{cases}$	10	
(e)	(i) Define Unit step function. (ii) Solve $y'' + 3y' + 2y = u(t-1) - u(t-2)$, $y(0) = 0$, $y'(0) = 0$.	10	
Q3	Attempt any four		
(a)	State and prove Linearity property of Laplace transform.	05	CO BT
(b)	If $\mathcal{L}[f(t)] = F(s)$ and c is any positive constant, Show that $\mathcal{L}[f(ct)] = \frac{1}{c}F(s/c)$.	05	CO BT
(c)	Find the inverse Laplace transform of $F(s) = \frac{2}{s^4} - \frac{48}{s^6}$	05	CO BT
(d)	Find inverse Laplace transform of $F(s) = \frac{-s+11}{s^2-2s-3}$	05	CO BT
(e)	Evaluate the line integral $\int_C \cos z \, dx + e^x \, dy + e^y \, dz$, where $C(t) = (x(t), y(t), z(t)) = (1, t, e^t)$ and $0 \leq t \leq 2$.	05	CO BT

[BT1: Remembering, BT2: Understanding, BT3: Applying, BT4: Analysing, BT5: Evaluating, BT6: Creating]

CO1. Develop the skills of Laplace Transforms, Fourier series and Fourier Transforms and their inverses.

CO2. Develop the skills of solving Partial differential equations.

CO3. Solve ODE's and PDE's using the properties of Laplace transform, Fourier series and Fourier Transforms.

CO4. Determine solutions of PDE for vibrating string and heat conduction.

CO5. Evaluate line integrals, surface integrals, and volume integrals and convert line integrals into area integrals and surface integrals into volume integrals using integral theorems.



**Shri Guru Gobind Singhji Institute of Engineering and Technology,
Nanded**

Department of BSH&M (Mathematics)

Summer term Examination Academic Year: 2024-25

Class: SY B.Tech.	Subject: M-III (Numerical methods and Differential equation).	Subject Code: BSC272
Date: 17.06.2025	Time: 14.30-17.30	Max. Marks: 100

Note: i) Nonprogrammable Calculator is allowed, ii) This paper contains total two pages.

Note: 1) Nonprogrammable Calculator is allowed, 2) Time 1 hr

Q.1	Attempt any six		Marks	CO	BT												
a	Using Bisection method determine a real root of the equation $f(x) = x^3 - x - 1 = 0$ lying between 1 and 1.5 (correct to two places of decimal).		07	CO2	BT3												
	Find the real root of the equation $xe^x - 3 = 0$ lying between 1 & 2, using regula falsi method correct to three decimal places.		07	CO2	BT3												
	By using Newton-Raphson's Method, find the real root of the equation $x^4 - x - 10 = 0$, correct to three places of decimals (you can take initial approximation as $x_0 = 2$).		07	CO2	BT3												
	Prove that, i) $\nabla = 1 - E^{-1}$ iii) $E\nabla = \Delta$ ii) $\delta = E^{\frac{1}{2}} - E^{-\frac{1}{2}}$ iv) $\Delta = \delta E^{\frac{1}{2}}$		07	CO2	BT2												
	Find $f(0.3)$ from the following given data: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>$f(x)$</td><td>3</td><td>6</td><td>11</td><td>18</td><td>27</td></tr></table>		x	0	1	2	3	4	$f(x)$	3	6	11	18	27	07	CO2	BT3
	x	0	1	2	3	4											
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For a certain town the population is given below. Find the population of 1953 by using Newton backward difference formula. <table><tr><td>Year</td><td>1931</td><td>1941</td><td>1951</td><td>1961</td></tr><tr><td>Populations in thousands</td><td>172</td><td>171.7</td><td>157.2</td><td>183.9</td></tr></table>		Year	1931	1941	1951	1961	Populations in thousands	172	171.7	157.2	183.9	07	CO2	BT3			
Year	1931	1941	1951	1961													
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Find the polynomial $f(x)$ of degree 2 such that $f(0) = 12$, $f(3) = 6$, $f(4) = 8$ by using Lagrange's interpolation formula.		07	CO2	BT3													
Q.2 Attempt any five																	
a	Given $y' = \frac{y-x}{y+x}$ with $y(0) = 1$, find $y(0.4)$ in 4 steps by Euler's method.		06	CO4	BT3												
	Evaluate $\int_0^6 \frac{1}{x^2+1} dx$ by using Simpson's $\frac{1}{3}$ rule with $h = 1$.		06	CO3	BT3												
	Find $\int_0^2 e^{x^2} dx$ approximately using trapezoidal rule with $h = \frac{1}{3}$.		06	CO3	BT3												

	d	Using Taylor's series, find $y(0.1)$ correct to four decimal places if $y(x)$ satisfies $\frac{dy}{dx} = x - y^2$ & $y(0) = 1$.	06	CO4	BT3										
	e	Given $\frac{dy}{dx} = 1 + y^2$ with $y(0) = 0$, find $y(0.2)$ by using Runge-kutta fourth order formula.	06	CO4	BT3										
	f	Given $\frac{dy}{dx} = x + y$ with initial condition $y(0) = 1$, find $y(0.05)$ by using Euler's Modified method (with $h = 0.05$) correct to two decimal places.	06	CO4	BT3										
Q.3 Attempt all															
	a	Find $\int_0^6 \frac{e^x}{1+x} dx$ approximately using Simpson's $\frac{3}{8}$ rule with $h = 1$.	06	CO3	BT3										
	b	Find the cubic polynomial which takes the following values: <table border="1"><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>$f(x)$</td><td>1</td><td>2</td><td>1</td><td>10</td></tr></table>	x	0	1	2	3	$f(x)$	1	2	1	10	06	CO1	BT3
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	a	Find the Fourier series of the periodic function $f(x) = \begin{cases} 2 & \text{if } -\pi \leq x < 0 \\ x & \text{if } 0 \leq x < \pi \end{cases}$ with $f(x + 2\pi) = f(x)$.	08	CO5	BT2										
	b	Find the Fourier cosine series of the periodic function $f(x) = x^2, -\pi < x < \pi$ with $f(x + 2\pi) = f(x)$.	06	CO5	BT2										
	c	Make only the difference table for the following data <table border="1"><tr><td>x</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>$f(x)$</td><td>31</td><td>69</td><td>131</td><td>223</td></tr></table>	x	3	4	5	6	$f(x)$	31	69	131	223	02	CO2	BT1
x	3	4	5	6											
$f(x)$	31	69	131	223											

Bloom's Taxonomy (BT): BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Develop the numerical skills for error analysis.
CO 2	Find roots of polynomial and transcendental equations using numerical techniques
CO 3	Evaluate numerical integration and differentiation.
CO 4	To use numerical methods to solve ordinary and partial differential equations and other engineering problems.
CO 5	Develop the skills of solving Partial differential equations using separation of variables and Fourier series.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded.

Summer Term Examination, 2024-25

S.Y. B. Tech.

Department: BSH&M (Mathematics)

Paper: Mathematics -III
(Numerical Methods, Probability and Statistics)

Course Code: BSC205

Date: 17/06/2025

Time: 14:30 - 17:30

Max. Marks: 100

Note: (i). Attempt all questions

(ii). Assume suitable data if necessary

(iii). Non-programmable calculators are allowed.

Q1. Attempt any five.		Marks	CO/ BT																
(a)	Find a real root of the equation $x^3 - x - 1 = 0$, lying between 1 and 2 correct upto two decimal places by using Bisection method.	08	CO1 BT3																
(b)	Find a real root of the equation $xe^x - 3 = 0$, lying between 1 and 1.5 correct upto four decimal places by using Regula-Falsi method.	08	CO1 BT3																
(c)	Find the negative real root of the equation $x^2 + 4 \sin x = 0$, by using the Newton-Raphson method.	08	CO1 BT3																
(d)	Prove that (i) $\Delta = E - 1$, (ii) $\nabla = 1 - E^{-1}$, (iii) $E = e^{hD}$, (iv) $\Delta = E\nabla = \nabla E$.	08	CO1 BT3																
(e)	Find the value of $\log_{10} 337.5$ from the following data: <table><tr><td>x</td><td>310</td><td>320</td><td>330</td><td>340</td><td>350</td><td>360</td></tr><tr><td>$f(x)$</td><td>2.4913</td><td>2.5051</td><td>2.5185</td><td>2.5314</td><td>2.5440</td><td>2.5563</td></tr></table>	x	310	320	330	340	350	360	$f(x)$	2.4913	2.5051	2.5185	2.5314	2.5440	2.5563	08	CO1 BT3		
x	310	320	330	340	350	360													
$f(x)$	2.4913	2.5051	2.5185	2.5314	2.5440	2.5563													
(f)	Find the value of $f(42)$ from the following table. <table><tr><td>x</td><td>20</td><td>25</td><td>30</td><td>35</td><td>40</td><td>45</td></tr><tr><td>$f(x)$</td><td>354</td><td>332</td><td>291</td><td>260</td><td>231</td><td>204</td></tr></table>	x	20	25	30	35	40	45	$f(x)$	354	332	291	260	231	204	08	CO1 BT3		
x	20	25	30	35	40	45													
$f(x)$	354	332	291	260	231	204													
Q2. Attempt any five.																			
(a)	Find the form of the function given below: <table><tr><td>x</td><td>3</td><td>2</td><td>1</td><td>-1</td></tr><tr><td>$f(x)$</td><td>3</td><td>12</td><td>15</td><td>-21</td></tr></table>	x	3	2	1	-1	$f(x)$	3	12	15	-21	08	CO2 BT3						
x	3	2	1	-1															
$f(x)$	3	12	15	-21															
(b)	Using following table find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x = 1.1$. <table><tr><td>x</td><td>1.0</td><td>1.1</td><td>1.2</td><td>1.3</td><td>1.4</td><td>1.5</td><td>1.6</td></tr><tr><td>y</td><td>7.989</td><td>8.403</td><td>8.781</td><td>9.129</td><td>9.451</td><td>9.750</td><td>10.031</td></tr></table>	x	1.0	1.1	1.2	1.3	1.4	1.5	1.6	y	7.989	8.403	8.781	9.129	9.451	9.750	10.031	08	CO2 BT3
x	1.0	1.1	1.2	1.3	1.4	1.5	1.6												
y	7.989	8.403	8.781	9.129	9.451	9.750	10.031												

(c)	Let $\frac{dy}{dx} = x^2(1+y)$ and $y(1) = 1, y(1.1) = 1.233, y(1.2) = 1.548, y(1.3) = 1.979$. Evaluate $y(1.4)$ by Adam-Bashforth method.	08	CO3 BT3																
(d)	Given $\frac{dy}{dx} = \frac{y-x}{y+x}$ with $y(0) = 1$, Using Euler's method find y for $x = 0.1$ in 4 steps.	08	CO3 BT3																
(e)	Using Runge-Kutta method, solve $\frac{d^2y}{dx^2} = y'' = (x^2 + y^2)(1 + y'^2)$ for $x = 0.5$ correct to four decimal places, subject to the initial conditions $y(0) = 1$ and $y'(0) = 0$ (Take $h = 0.5$).	08	CO3 BT2																
(f)	Solve $y' = 2e^x - y$ at $x = 0.4$ and $x = 0.5$ by Milne's method, give their values at the four points. <table border="1" data-bbox="496 674 986 761"><tr><td>x</td><td>0</td><td>0.1</td><td>0.2</td><td>0.3</td></tr><tr><td>y</td><td>2</td><td>2.010</td><td>2.040</td><td>2.090</td></tr></table>	x	0	0.1	0.2	0.3	y	2	2.010	2.040	2.090	08	CO3 BT3						
x	0	0.1	0.2	0.3															
y	2	2.010	2.040	2.090															
Q3. Attempt any four.																			
(a)	Using Taylor's series for $y(x)$, find $y(0.1)$ correct to four decimal places if $y(x)$ satisfies $\frac{dy}{dx} = y' = x - y^2, y(0) = 1$ where $x_0 = 0$.	05	CO3 BT3																
(b)	Find the value of the integral $\int_0^6 \frac{dx}{1+x^2}$ by using Trapezoidal rule for $h = 1$.	05	CO2 BT2																
(c)	Estimate the value of the integral $\int_0^4 e^x dx$ by using Simpson's $\frac{1}{3}$ rule with $h = 1$.	05	CO2 BT3																
(d)	The table below shows the scores (out of 60) of the applicants in an aptitude test. Calculate the mean. <table border="1" data-bbox="237 1207 1281 1285"><tr><td>Scores</td><td>1-10</td><td>11-20</td><td>21-30</td><td>31-40</td><td>41-50</td><td>51-60</td></tr><tr><td>No. of applicants</td><td>5</td><td>6</td><td>8</td><td>10</td><td>7</td><td>4</td></tr></table>	Scores	1-10	11-20	21-30	31-40	41-50	51-60	No. of applicants	5	6	8	10	7	4	05	CO4 BT2		
Scores	1-10	11-20	21-30	31-40	41-50	51-60													
No. of applicants	5	6	8	10	7	4													
(e)	Find out the missing frequencies in the following distribution if $N = 100$ and median is 32 <table border="1" data-bbox="237 1359 1272 1438"><tr><td>Marks</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td><td>50-60</td><td>Total</td></tr><tr><td>No. of students</td><td>10</td><td>f_2</td><td>25</td><td>30</td><td>f_5</td><td>10</td><td>100</td></tr></table>	Marks	0-10	10-20	20-30	30-40	40-50	50-60	Total	No. of students	10	f_2	25	30	f_5	10	100	05	CO4 BT2
Marks	0-10	10-20	20-30	30-40	40-50	50-60	Total												
No. of students	10	f_2	25	30	f_5	10	100												

[BT1: Remembering, BT2: Understanding, BT3: Applying, BT4: Analysing, BT5: Evaluating, BT6: Creating]

CO1. Find roots of polynomial and transcendental equations using numerical techniques

CO2. Evaluate numerical integration and differentiation.

CO3. To use numerical methods to solve ordinary and partial differential equations and other engineering problems.

CO4. Develop problem solving techniques needed to accurately calculate probabilities.

CO5. Describe problems properties of discrete and continuous distribution functions and its applications to solve problem.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of BSH&M (Mathematics)
Summer term Examination Academic Year: 2024-25 (Semester – I)

Class: SY B.Tech.	Subject: M-III (Complex analysis, Probability and Statistics).	Subject Code: BSC203
Date: 17/06/2025	Time: 14.30-17.30	Max. Marks: 100

Note: i) Nonprogrammable Calculator is allowed, ii) This paper contains total 2 page.

Q.1	Attempt any five	Marks	CO	BT
a	i) Find all the 4 th roots of $z = -1 + \sqrt{3}i$. ii) Sketch the set of solutions of following inequality $ Arg(z) \leq \frac{\pi}{4}$.	06	CO1	BT1
b	i) Find modulus and argument of $z = (1 + \sqrt{3}i)^{30}$. ii) Find $Re(f)$ & $Im(f)$ for $f(z) = \frac{z}{z+1}$.	06	CO1	BT3
c	Find the points at which the function $f(z) = e^{-y}(\cos x + i \sin x)$ is analytic.	06	CO1	BT2
d	Find harmonic conjugate of the function $u(x, y) = x^2 - y^2 - x$.	06	CO1	BT2
e	i) Find the principal logarithm (Lnz) for the complex number $z = \sqrt{3} - i$. ii) Find the principal value of $(1 - i)^{1+i}$ (Write solution in detail).	06	CO1	BT1
f	i) Find all the points at which the function $f(z) = z(z^4 - 5)$ is not conformal. ii) Find the fixed points of following function $f(z) = \frac{2z-5}{z+4}$.	06	CO2	BT2
Q.2	Attempt any five			
a	i) Evaluate $\int_C \frac{z^3 e^z}{(z-2)^2} dz$ where C is the circle of radius 5 with centre at 0 oriented anticlockwise. ii) Find $\int_C \frac{\cos(z^2+3)}{e^z} dz$, where C is the triangle with vertices $i, -1-i$ & $1-i$ with anticlockwise orientation.	06	CO3	BT3
b	Evaluate $\int_C (2x + 1 + iy) dz$, where C is the path given by $c(t) = (t + 1) + i(2t^2 - 1), 0 \leq t \leq 1$.	06	CO3	BT2
c	10 dissimilar balls are distributed at random into 4 boxes marked A, B, C, D. Find the probability that these boxes contain respectively 4, 3, 2, 1 balls.	06	CO5	BT2
d	Five men in a company of 15 are graduates. If three men are picked out of the 15 at random then i) what is the probability that they are all graduates? ii) what is the probability of exactly one is graduate?	06	CO5	BT2
e	A coin is tossed 10 times. Find the probability of getting (i) Exactly 7 heads (ii) 3 heads and 7 tail (iii) At most 2 heads.	06	CO5	BT2

	f	Three boxes I, II and III, contain balls as follows: • Box I contains 2 white & 4 black balls. • Box II contains 3 white & 1 black balls. • Box III contains 2 white & 3 black balls. A box is chosen at random, and a ball is drawn out of it. If the ball is found to be black, what is the probability that Box III was selected.	06	CO5	BT3																
Q.3 Attempt any five																					
	a	A box contains 6 white and 4 black balls. If 3 balls are drawn at random, Find the mathematical expectation of the number of white balls.	06	CO5	BT2																
	b	Given that $P(A) = \frac{5}{8}$, $P(B) = \frac{3}{8}$ & $P(A \cup B) = \frac{3}{4}$, Find the probabilities: $P(\bar{A})$, $P(A \cap B)$, $P(A B)$, $P(\bar{A} \cap B)$.	06	CO5	BT1																
	c	The p.d.f. of a continuous random variable X is $y = k(-x^2 + 3x - 2)$; $(1 \leq x \leq 2)$. Determine, i) The value of the constant k. ii) The cumulative distribution function. iii) The probability that X is less than $\frac{3}{2}$.	06	CO5	BT2																
	d	The following is the table which gives you the distribution of marks secured by some students in an examination. Find the median for the following data. <table><tr><td>Marks</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td><td>50-60</td></tr><tr><td>No. of students</td><td>42</td><td>85</td><td>120</td><td>84</td><td>49</td></tr></table>	Marks	10-20	20-30	30-40	40-50	50-60	No. of students	42	85	120	84	49	06	CO5	BT2				
Marks	10-20	20-30	30-40	40-50	50-60																
No. of students	42	85	120	84	49																
	e	Find the Mean and mode of the following distribution <table><tr><td>Class</td><td>20-30</td><td>30-40</td><td>40-50</td><td>50-60</td><td>60-70</td></tr><tr><td>Frequency</td><td>7</td><td>20</td><td>8</td><td>10</td><td>5</td></tr></table>	Class	20-30	30-40	40-50	50-60	60-70	Frequency	7	20	8	10	5	06	CO5	BT1				
Class	20-30	30-40	40-50	50-60	60-70																
Frequency	7	20	8	10	5																
	f	i) Determine the value of "a" for which the function $u(x, y) = e^{3x} \cos(ay)$ is harmonic. ii) If $z = \cos(3\pi i)$ then find $Re(z)$.	06	CO1	BT2																
Q.4 Attempt all																					
	a	The marks obtained by 10 students are: <table><tr><td>Marks (x)</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td></tr><tr><td>Frequency (f)</td><td>1</td><td>2</td><td>3</td><td>2</td><td>2</td></tr></table> Find the Mean ($\bar{x}$) & standard deviation(σ) for the above data.	Marks (x)	2	4	6	8	10	Frequency (f)	1	2	3	2	2	06	CO1	BT1				
Marks (x)	2	4	6	8	10																
Frequency (f)	1	2	3	2	2																
	b	A discrete random variable X has the following probability mass function: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>f(x)</td><td>k</td><td>3k</td><td>5k</td><td>7k</td><td>9k</td><td>11k</td><td>13k</td></tr></table> i) Find value of k ii) Find $P(X < 4)$, $P(X \geq 5)$ & $P(3 < X \leq 6)$.	x	0	1	2	3	4	5	6	f(x)	k	3k	5k	7k	9k	11k	13k	04	CO1	BT2
x	0	1	2	3	4	5	6														
f(x)	k	3k	5k	7k	9k	11k	13k														

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

	On successful completion of this course, students should be able to:
CO 1	Determine whether a given function is analytic and apply analyticity on harmonic functions and conjugates of harmonic functions.
CO 2	Transform a region to another region using conformal mapping.
CO 3	To evaluate contour integrals using Cauchy's integral theorem and formula.
CO 4	Represent functions as Taylor, power, and Laurent series, classify singularities and poles, find residues and evaluate complex integrals using the residue theorem.
CO 5	Develop problem solving techniques needed to accurately calculate probabilities and describe the properties of discrete and continuous distribution functions.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Information Technology
Summer Term Examination
Academic Year: 2024-25 (Semester – II)

Class: SY	Subject: Discrete Mathematics	Subject Code: PCC-IT202/202
Date: 25.06.2025	Time: 14:30	Max. Marks: 100

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Solve every question step by step and Write proper Formula/ Theorem if question based on that.

Q.N.		CO	BT
1	Show that the relation $R = \{ (a,b) \mid a, b \in \mathbb{Z}, a - b \text{ divisible by } 3 \}$ is an equivalence relation.	1	3
2	Check given diagrams is lattice or not (A) (B)	1	3
3	Suppose repetition is not permitted. a) how many 4-digit number can be formed, six digits 1, 2, 3, 5, 7, 8? b) How many in a) such numbers are less than 4000? c) How many in a) such numbers are even? d) How many in b) such numbers are odd? e) How many in a) such numbers contain both 3 and 5?	4	4
4	A company imported bolt from 3 different company A, B, C. From A 30%, from B 40% and from C 30% bolt imported but there are some defective bolts are there such as A has 2% defective, B has 3% defective and C has 5% defective bolt. If a random bolt is selected from all combination what is the probability a) that it was manufactured by the company A b) that it was manufactured by the company B c) that it was manufactured by the company C	4	4
5	Determine the following tautology or contradiction or contingency $[(p \vee q) \rightarrow (r \wedge s)] \leftrightarrow [\neg(p \vee q) \vee (r \wedge s)]$	2	3
6	Let $R = \{0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 360^\circ\}$ and $*$ is a binary operation so that a and b in R, $a*b$ is overall angular rotation corresponding to successive rotation by a and b. show that $(R, *)$ is a group. OR If set Q of all rational number other than 1 with $a*b = a+b-ab$. Show that $(Q, *)$ is a group. Also check $(Q, *)$ is abelian group or not	3	4

7.	Explain Eulerian path and Hamiltonian path in detail with examples. Also check given graphs (A, B) is Eulerian or Hamiltonian? A B	5	2,4
8	For the following set weights construct optimal binary prefix code using a Huffman Tree algorithm and find the optimal binary prefix code for each character and average bits/char. $a=45, b=3, c=30, d=2, e=5, f=15$. Also find character code for below sequence "011011111000011011001001111100"	5	3,4
9	Explain what minimum spanning tree (MST) with the help of Kruskal's and Prim's algorithm. Also find MST for the given graph	5	2, 4
10	Write Breadth First Traversal (BFT) algorithm with example and crate 4 different BFT for given graph.	5	2, 4

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	For a given logic sentence express it in terms of predicates, quantifiers, and logical connectives.
CO 2	For a given a problem, find the solution using deductive logic and prove the solution based on logic.
CO 3	For a given a mathematical problem, classify its algebraic structure.
CO 4	Evaluate Boolean functions and simplify expressions using the properties of Boolean algebra.
CO 5	Develop the given problem as graph networks and solve with techniques of graph theory.